Team Details

- ➔ Team name - ARCANE
- ➔ Team leader name – Abhirama Krishna P
- ➔ Problem Statement - The Challenge – Fragmented, Reactive Sanitation & Climate Adaptation

Brief about the idea

Things are getting really bad with urban sanitation around the world in 2026. I mean, its turned into this huge crisis especially in big cities that are growing fast. Over half of the 100 largest ones are dealing with high water stress or even worse, based on what UN Habitat says lately.

That idea of a global water bankruptcy era, that's what UN scientists are calling it now. It comes from all this damage to rivers and underground water sources that can't be fixed easily. Urbanization is pushing it, along with changing climate patterns and not enough infrastructure to handle things. So many people, like 2.5 billion, don't have safe ways to manage sanitation. And that Sustainable Development Goal 6, its way behind schedule because of all this.

In those informal settlements where one in eight city people live, the problems hit hardest. Water gets contaminated and leads to diseases spreading, you know like dengue or cholera outbreaks. Then there's sewage backing up in heavy rain, flooding streets and even inside homes. Sanitation workers have it rough too, going into tight spaces without proper air and getting exposed to toxins, since they're not equipped right.

Cities like Lagos, Manila, or Mumbai show how it plays out. They have data that's all over the place, just complaints and old manual records, so maintenance is only reactive through hotlines. That causes delays and mix ups, even doing the same fixes twice. In those crowded low-income areas, it just makes health risks worse for everyone.

Half of the water needs in cities could jump 50 to 70 percent by the middle of the century. But investments aren't keeping up at all. Even normal storms turn into big disasters now, displacing millions of people and hurting the economy. Inequality gets deeper too, especially for groups like women and kids who end up fetching unsafe water or dealing with open defecation most of the time. It seems like that part stands out, how it affects them unevenly. I think the whole thing is messy and not fully sorted out yet.

Opportunities

How different is it from any of the other existing ideas?

- Focuses on predictive, proactive AI forecasting (24–72h overflow risks) vs. mostly reactive hardware like robotic cleaners (Bandicoot, R-Bot/Solinas, Swasth AI).
- Integrates multi-source data (citizen reports, rainfall APIs, municipal logs, satellite) for city-wide visibility and task dashboards, unlike isolated robotic cleaning or sensor-based hygiene monitoring (e.g., Indore toilets, Coimbatore AI).
- No mainstream tool combines real-time grid/maps, predictive overflow/blockage forecasting, automated task assignment, and worker hazard alerts in one unified platform.
- Extends to rural side with 5–10 year climate risk projections + optimized adaptations (e.g., crop shifts, shade nets), beyond weather alerts or macro forecasts.
- Unique "nervous system" approach: AI as brain for prevention across gritty street assets (drains/manholes/toilets) + urban-rural synergies, mobile-first SaaS — not just robots/sensors.

Opportunities

How will it be able to solve the problem?

- Breaks data silos by unifying scattered sources (IoT, citizen inputs, APIs, logs) to end WhatsApp/paper chaos and provide holistic visibility.
- Enables predictive/proactive maintenance via AI (LSTM/time-series) for 24–72h forecasts, preventing overflows, flooding, diseases, and worker hazards (toxic gases).
- Enhances coordination/safety with color-coded maps, drag-drop task assignment, mobile alerts/checklists — reduces delays, duplicates, risks.
- Supports rural planning with long-term risk projections + budget/labor-optimized adaptation recommendations for panchayats.
- Scales equitably (mobile-first, low-bandwidth, low-income focus), lowers costs (preventive > emergency), cuts health/climate impacts — turns unpredictable failures into preventable events.

USP of the proposed solution

- First-mover in sanitation-specific predictive AI for street-level assets in emerging cities ("urban sanitation nervous system").
- Dual urban-rural integration in one hub, with synergies (e.g., drainage data informing flood models).
- End-to-end prevention workflow: monitoring + forecasting + alerts + automation + optimization + reports — beyond robots or basic sensors.
- Equity/feasibility emphasis: prioritizes vulnerable areas, uses low-cost/open data, mobile-first React app, scalable SaaS potential.
- High human-centered impact: reduces worker deaths, flooding/diseases, crop losses — strong public health, safety, resilience ROI, TN-tailored but globally replicable.

List of features offered by the solution

Urban Sanitation Module Features

Real-time asset cards in grid view
- -Fill level % (color-coded: green <50%, yellow 50–80%, red >80%)
- -Blockage risk score (0–100)
- -Current flow status (litres per second, with trend arrows)
- -Asset type (Manhole / Open Drain / Community Toilet)
- -Location name, landmark, and coordinates

Interactive map view (Leaflet-based)
- -Chennai zone overlay (e.g., Zones 9–13)
- -Coloured pins for each asset
- -Heatmap highlighting priority/high-risk areas
- -Zoom, pan, and layer toggle controls

Critical Alerts banner
- -Prominent red/orange warning for high-risk assets
- -Lists affected asset IDs and urgency message
- -Example: “High overflow risk detected in 120, 178. Immediate action required.”

Predictive analytics
- -24-hour overflow probability (displayed as % in bold red when >50–80%)
- -Trend line chart showing fill/flow progression over last 24 hours

Task assignment & management
- -“+ Assign Maintenance Crew” button on asset details
- -Active Tasks panel showing assigned jobs with:
  - Type (CLEANING, REPAIR, etc.)
  - Priority (CRITICAL / HIGH)
  - Status & time assigned
  - Notes (e.g., “Severe blockage reported by sensor”)

Worker safety alerts
- -Mobile push-style notifications for hazards (toxic gas, sudden overflow)
- -Quick safety checklist access

List of features offered by the solution

Rural Climate Adaptation Module Features

Village-specific interactive risk map (Leaflet)
- -Layered visualization: flood exposure polygons, heat vulnerability gradient, crop failure zones.
- -Toggle between risk types
- -Centred on Tamil Nadu districts with zoom functionality

5–10 year climate risk projections
- Bar charts comparing current vs. 5 years vs. 10 years for flood, heat, crop failure
- Radar/vulnerability diagram showing multi-hazard exposure

Top Recommendations engine
- -Ranked adaptation strategies based on constrained optimization
- -Details per recommendation:
  - Name (e.g., Heat-Reflective Shade Nets)
  - Primary crop targeted (e.g., Jasmine, Millet)
  - Estimated cost (e.g., ₹30,000)
  - ROI percentage (e.g., 75%)
  - Impact score (e.g., 60/100)
  - Short description of benefit
- -“Add to Plan” button

Report generation
- -“Generate Full Report” button → exports tailored PDF with map, projections, and selected recommendations

Cross-Platform & General Features

Role-based dashboards
- -Municipal Official: full urban + rural overview, assignment tools
- -Sanitation Worker: mobile-optimized alerts, task list, hazard warnings
- -Panchayat Member: focused rural risk maps & recommendations

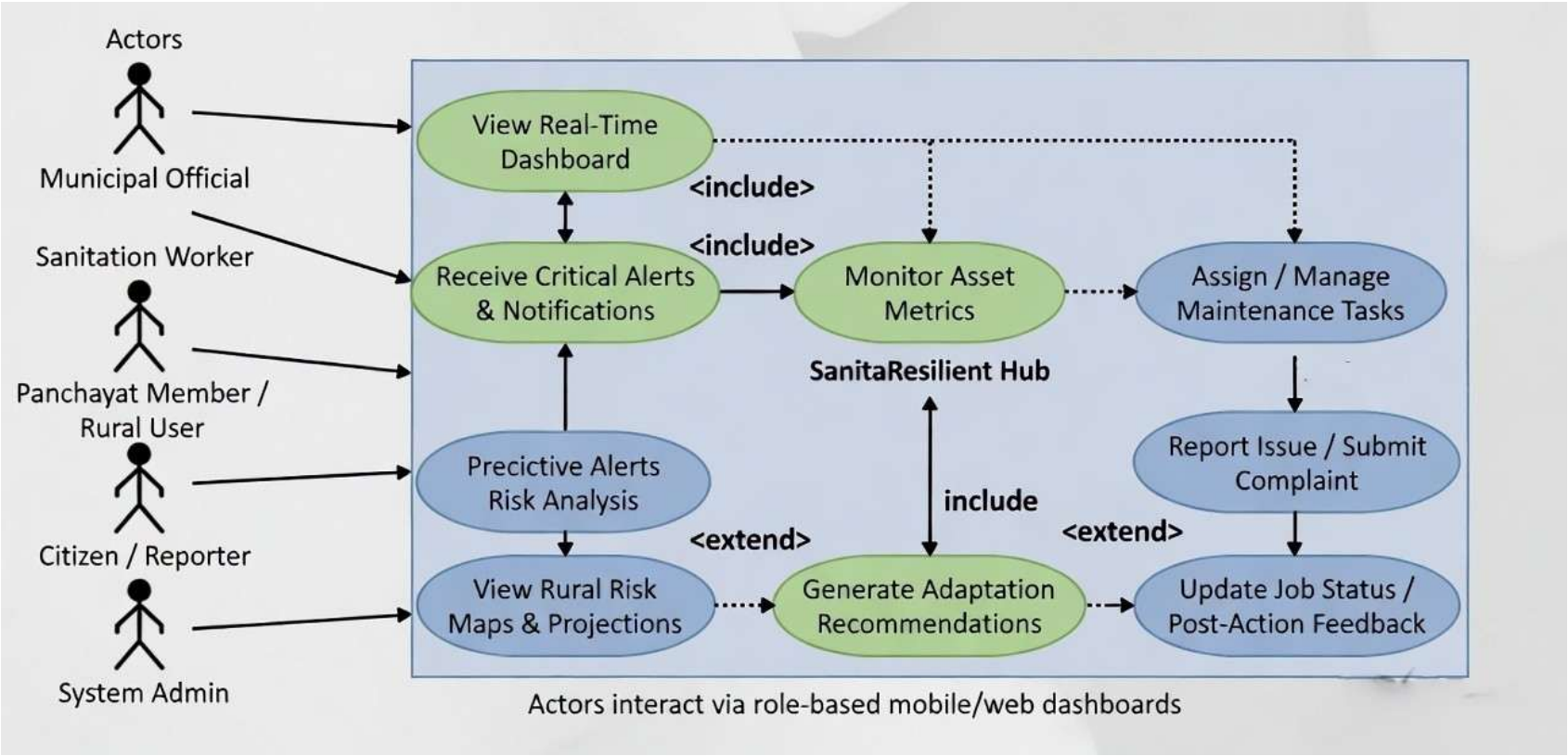
Real-time updates
- -Simulated or live sensor data refresh (every 10–30 seconds in demo)
- -Socket.io-powered live pushes for alerts and metric changes

Mobile-first responsive design
- -Optimized for low-bandwidth Android phones used in Chennai streets and TN villages

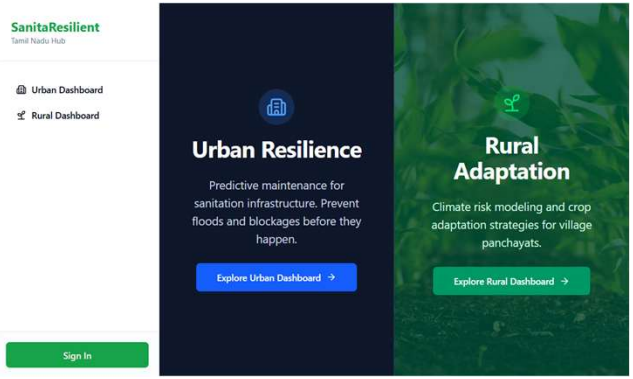
User interface elements
- -Earthy colour scheme (greens for safe, blues for flow, reds/oranges for alerts)
- -Sidebar navigation (Urban Dashboard, Rural Dashboard, Logout)
- -User profile display (name, role, e.g., “Ravi Kumar – MUNICIPAL OFFICIAL”)

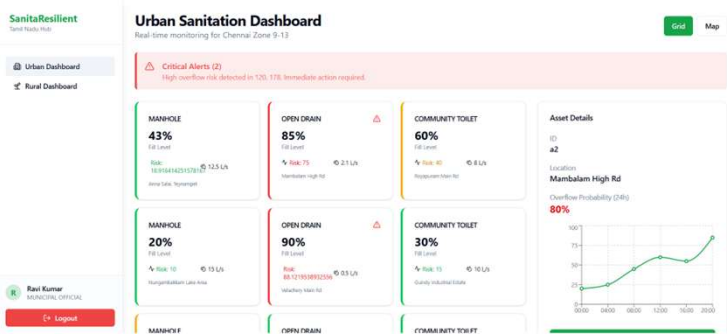
Prototype-ready tech stack highlights
- -Built with React, Vite, TypeScript, Tailwind CSS
- -Visuals: Recharts (gauges, lines), React-Leaflet (maps)
- -Data handling: TanStack Query, Axios, Socket.io
- -Mock data for Chennai zones & TN villages during development

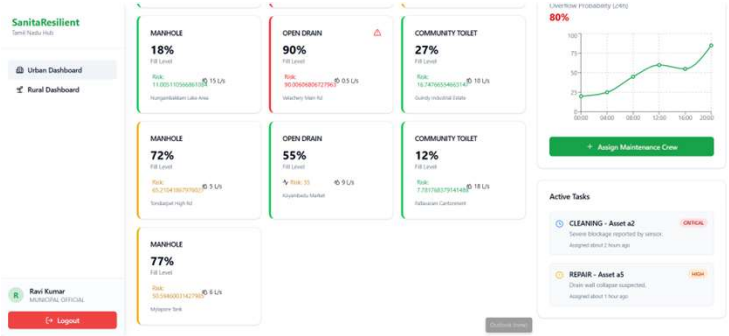
Use-case diagram

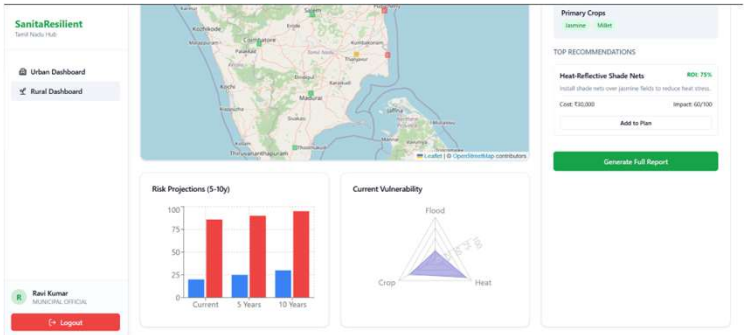


Wireframes/Mock diagrams of the proposed solution

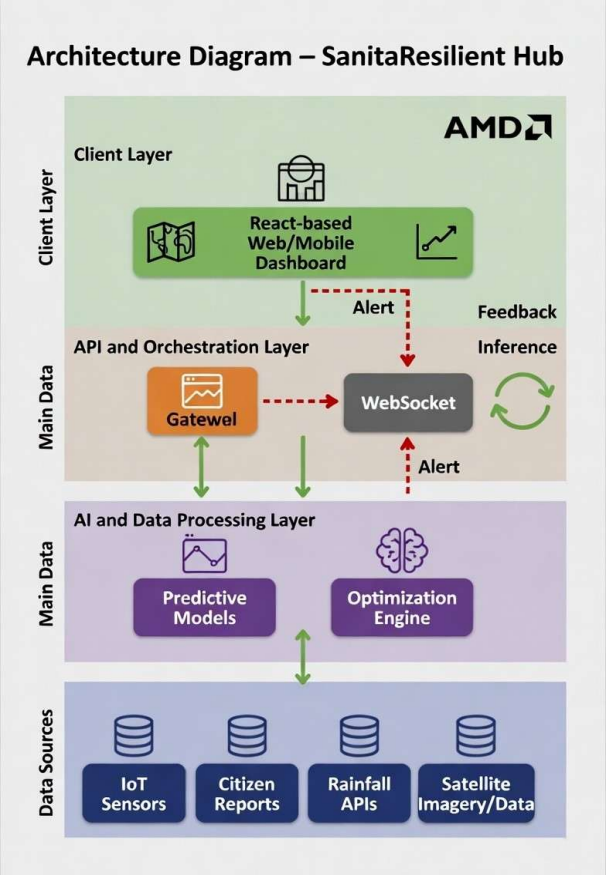








Architecture diagram of the proposed solution



Technologies to be used in the solution

- Core Web/App: React, Vite, TypeScript, Tailwind CSS, shadcn/ui, Lucide icons
- Charts/Maps: Recharts, React-Leaflet + Leaflet
- State & Queries: Zustand, TanStack Query
- Real-time: Socket.io-client
- AI/ML (backend scripts): Python, LSTM/Prophet/XGBoost (forecasting), PuLP (optimization), GeoPandas/Rasterio
- Data: PostgreSQL/PostGIS (planned), mock data + public APIs for prototype
- Deployment: Vercel/Netlify (frontend), Node.js/Express (API prototype)

Usage of AMD Products/Solutions

AMD Hardware/Software Relevance

- AMD Ryzen™ AI / Ryzen Embedded (edge inference for low-latency predictions on municipal servers or field gateways)
- AMD Radeon™ graphics (accelerated rendering of maps & dashboards)
- AMD ROCm™ platform (AI model optimization & inference acceleration)

Prototype Assets (Optional)

- ➔ GitHub Public Repository Link
- ➔ Demo Video Link (Max: 3 Minutes)

Add as per the requirements of the contest

Type here

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACK2SKILL

Thank you!